

Ethical Hacking Laboratory

Kali Linux Commands – Definitions and Syntax

Based on the uploaded LAB MANUAL: Ethical Hacking Laboratory – Kali Linux Commands and Nmap Scanning.

1. Basic Linux Commands

Command	Definition	Syntax
rm	Removes files and directories.	rm filename
users	Displays usernames of users currently logged in.	users
tree	Displays files and directories in a tree-like structure.	tree
ls	Lists files and directories.	ls
cat	Displays the contents of a file.	cat filename
mkdir	Creates a new directory.	mkdir directory_name
mv	Moves or renames files/directories.	mv source destination
touch	Creates an empty file.	touch filename
pwd	Displays the current working directory.	pwd
cd	Changes the current directory.	cd directory_name
cp	Copies files or directories.	cp source destination

2. File Viewing and Editing Commands

Command	Definition	Syntax
less	Displays file contents page by page.	less filename.txt
more	Displays file contents one screen at a time.	more filename.txt
vi	Text editor used to create and edit files.	vi filename.txt
head	Displays the first 10 lines of a file by default.	head filename.txt
tail	Displays the last 10 lines of a file by default.	tail filename.txt

3. System Information and Text Processing

Command	Definition	Syntax
uptime	Shows how long the system has been running.	uptime
date	Displays the current date and time.	date
free	Displays memory usage information.	free
free -h	Displays memory usage in human-readable format.	free -h
history	Displays previously executed terminal commands.	history
history 10	Displays the recent 10 commands.	history 10
sort	Sorts lines of a text file.	sort file.txt
sort -r	Sorts lines in reverse order.	sort -r file.txt

Command	Definition	Syntax
wget	Downloads files from the Internet.	wget URL

4. Nmap Commands

Command	Definition	Syntax
nmap	Scans a target for available ports/services.	nmap target-ip
nmap -sV	Detects service versions running on target ports.	nmap -sV target-ip
nmap -A	Performs OS detection, version detection, script scanning and traceroute.	nmap -A target-ip
nmap -O	Attempts to identify the operating system of the target.	sudo nmap -O target-ip
nmap -sS	Performs a TCP SYN scan.	nmap -sS target-ip
nmap -sn	Performs host discovery without a port scan.	nmap -sn network
nmap -p	Specifies the ports to scan.	nmap -p port-range target-ip

5. Nmap Timing Templates

Command	Definition	Syntax
-T0	Paranoid timing; very slow scan.	nmap -T0 target-ip
-T1	Sneaky timing; slow scan.	nmap -T1 target-ip
-T2	Polite timing; slows scanning.	nmap -T2 target-ip
-T3	Normal/default balanced timing.	nmap -T3 target-ip
-T4	Aggressive/faster timing.	nmap -T4 target-ip
-T5	Insane/very fast timing.	nmap -T5 target-ip

6. Password Auditing

Command	Definition	Syntax
hydra	Used in the manual for password auditing against an authorized test service.	hydra -l username -P password_file service://target-ip

7. Information Gathering

Command	Definition	Syntax
theHarvester	Gathers publicly available information about a domain.	theHarvester -d domain -b source
whois	Retrieves domain information records.	whois domain
site:	Google search operator for pages belonging to a specified domain.	site:domain
filetype:	Google search operator for a specified file type.	filetype:pdf site:domain

8. Windows Networking Commands

Command	Definition	Syntax
tracert	Traces the route to a destination host.	tracert hostname/IP

Command	Definition	Syntax
ping	Tests network connectivity to a host.	ping hostname/IP
ipconfig	Displays Windows IP configuration.	ipconfig
ipconfig /all	Displays detailed Windows network configuration.	ipconfig /all
netstat -an	Displays network connections and listening ports numerically.	netstat -an

9. Web Vulnerability Analysis

Command	Definition	Syntax
nikto	Used in the manual to identify web-server vulnerabilities.	nikto -h target-url

10. Packet Analysis

Command	Definition	Syntax
Wireshark	Graphical network packet analyzer used to capture and analyze network traffic.	Open Wireshark → select interface → start capture
tcpdump	Command-line tool used to capture/analyze network packets.	tcpdump -i interface

11. MSFVenom

Command	Definition	Syntax
msfvenom	Used in the manual to generate a sample payload for educational testing.	msfvenom -p payload-options LHOST=IP LPORT=PORT -f format > filename

12. Batch File Commands

Command	Definition	Syntax
@echo off	Prevents commands in a batch file from being displayed.	@echo off
echo	Displays text on the screen.	echo message
pause	Pauses batch-file execution.	pause
Batch execution	Executes a batch file.	filename.bat

13. Shell Programming

Command	Definition	Syntax
#!/bin/bash	Specifies Bash as the interpreter for the shell script.	#!/bin/bash
read	Reads user input into a variable.	read variable
Arithmetic	Performs arithmetic using Bash arithmetic expansion.	result=\$((expression))
if	Executes commands when a condition is true.	if [condition]\nthen\n commands\nfi
elif	Checks another condition if the previous condition is false.	elif [condition]\nthen\n commands
else	Executes commands when preceding conditions are false.	else\n commands

Important Commands for Viva/Exam

Linux: rm, users, tree, ls, cat, mkdir, mv, touch, pwd, cd, cp

File handling: less, more, vi, head, tail, sort

System: uptime, date, free, history

Nmap: nmap, -sV, -A, -O, -sS, -sn, -p, -T0 to -T5

Security tools: hydra, theHarvester, whois, nikto, tcpdump, msfvenom

Windows: ping, tracert, ipconfig, netstat